

WEEK 2 · SESSION 4

Ultrasonic & IMU

“How far?” and “Which way?”

Two new senses: the world ahead and the robot's orientation.

Sensing yourself vs. sensing the world

HUMANS · PROPRIOCEPTIVE

What can you sense in a sensory deprivation tank?

Sense limb position, joint movement, and muscle tension—even with your eyes closed.

HUMANS · EXTEROCEPTIVE

What is around me?

Vision, hearing, touch, smell, and taste reveal the outside world.

ROBOTS · PROPRIOCEPTIVE

Encoders: wheel and joint rotation.

Gyro + accelerometer: body motion and tilt, analogous to our inner-ear balance sense.

ROBOTS · EXTEROCEPTIVE

Range finders: distance to objects.

Cameras: objects and landmarks.

Contact sensors: touching a surface.



Living Without Proprioception

Paul Chrustowski



Watch on  YouTube

Two different questions



ULTRASONIC

How far away?

Measures distance to an object ahead.



IMU / GYRO

Which way am I facing?

Measures orientation and turning.

Environment + orientation are not the same as position on a map.

Ultrasonic: time an echo

1 · SEND

A short sound pulse leaves the sensor.

2 · BOUNCE

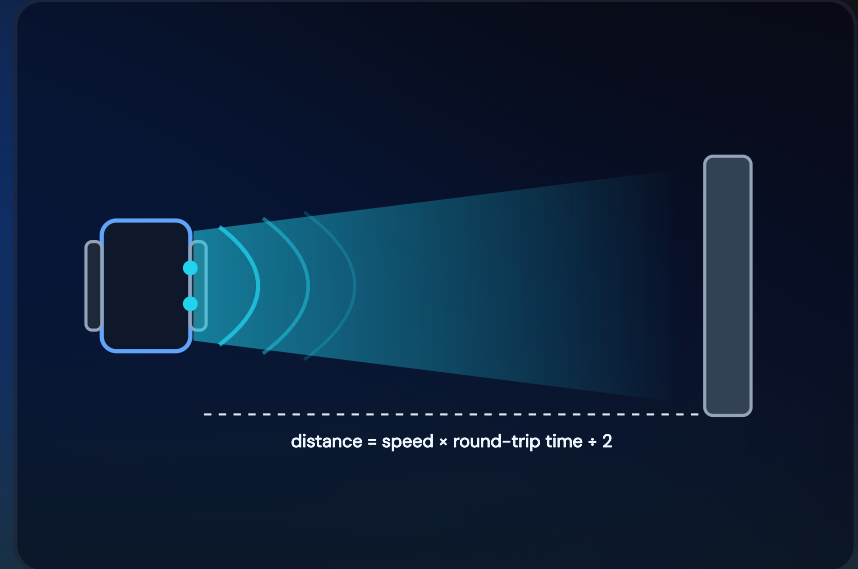
The pulse reflects from an object.

3 · RECEIVE

The return time reveals distance.

$$d = \frac{c \Delta t}{2}$$

Divide by 2 because the sound travels out and back.



Ultrasonic in the real world

BEST CASE

Flat, hard object directly in front.

ANGLED SURFACE

The echo reflects away from the sensor.

SOFT SURFACE

Fabric absorbs sound and weakens the return.

Noisy or missing readings are physical behavior—not automatically bad code.

Ultrasonic: failure modes → filters

1 · RANDOM NOISE

Small jitter around the true distance; often modeled as Gaussian noise.

Filter: average a few valid readings or smooth over time. More smoothing adds delay.

2 · NO RETURNING ECHO

XRP's XRPLib returns **65535** on timeout: an invalid-reading code, not a distance.

Filter: reject 65535 and values outside 2–400 cm before averaging. Mark missing data “unknown.”

3 · MULTIPLE REFLECTIONS

Extra bounces lengthen the echo path and can produce a sudden, overly large distance.

Filter: flag jumps too large for speed × time; recheck. A short median filter rejects isolated spikes.

4 · AN OBJECT ENTERS THE BEAM

A hand or passing object can abruptly replace the wall as the measured target.

Filter: for wall tracking, flag and recheck jumps. For obstacle avoidance, treat a new close object as real.

Same stationary target + nearly fixed beam: flag $|\Delta d| > v_{\max} \Delta t + \text{noise margin}$.
Turning or switching targets can cause real jumps—even when the robot moves slowly.

Gyroscope: how fast am I turning?

ANGULAR VELOCITY · ω

A gyroscope measures rotational velocity, usually in $^{\circ}/s$ or rad/s .

A 3-axis gyro reports rotation rates about the robot's x, y, and z axes.

Top view · level robot

$$\omega_z = +30^{\circ}/s$$



z points out of the screen

Positive turn: counterclockwise

Rate × time = change in position

STRAIGHT-LINE MOTION

$$\Delta x = v \Delta t$$

$$2 \text{ m/s} \times 3 \text{ s} = 6 \text{ m}$$

Constant velocity × elapsed time gives displacement.

ROTATION ABOUT ONE AXIS

$$\Delta \theta = \omega \Delta t$$

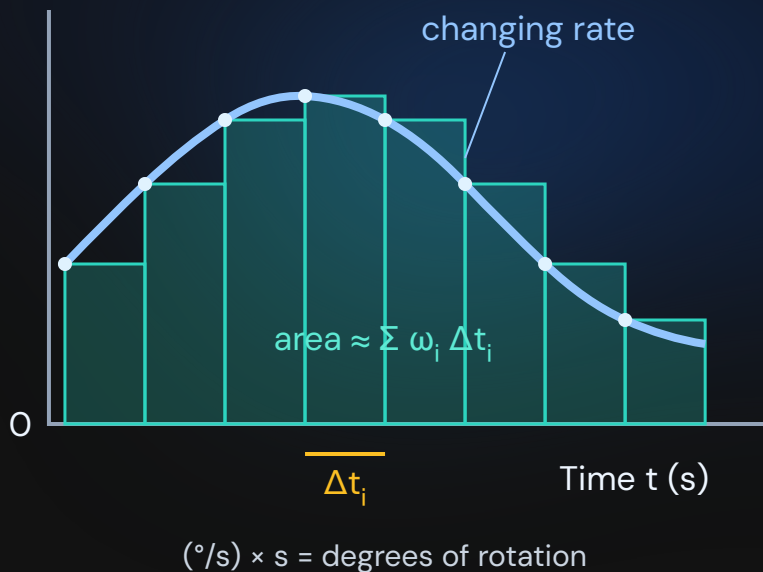
$$30^\circ/\text{s} \times 3 \text{ s} = 90^\circ$$

Constant angular velocity × elapsed time gives angle change.

Add the change to the starting angle: $\theta = \theta_0 + \Delta\theta$. Direction gives the sign.

Changing rate? Add tiny turns.

Angular velocity ω ($^\circ/\text{s}$)



A RIEMANN SUM

Sample rapidly. Treat the rate as constant over each short time interval.

Each rectangle has area $\omega_i \Delta t_i$: one tiny angle change.

Add the rectangles to approximate the area under the curve.

Signed area = angular displacement $\Delta\theta$.

The sum becomes a loop

MATH · ONE-AXIS ROTATION

$$\theta_N \approx \theta_0 + \sum_{i=0}^{N-1} \omega_i \Delta t_i$$

$$\Delta t_i = t_{i+1} - t_i$$

Each update adds one rectangle:

$$\theta \leftarrow \theta + \omega_i \Delta t_i$$

PYTHON · RECORDED SAMPLES

```
# t: timestamps in seconds
# omega: z-axis rates in deg/s
theta = 0.0 # starting angle, deg
for i in range(len(t) - 1):
    dt = t[i + 1] - t[i]
    theta += omega[i] * dt
```

Use measured elapsed time, even if the sampling interval varies.

A 0.5°/s bias adds 30° of error in 60 s. Integrated gyro angles drift.

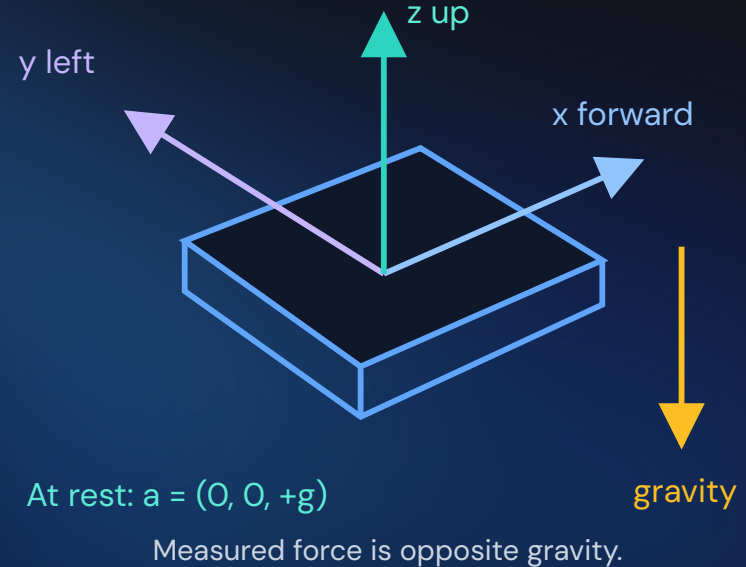
Accelerometer: three axes of acceleration

READINGS · a_x , a_y , a_z

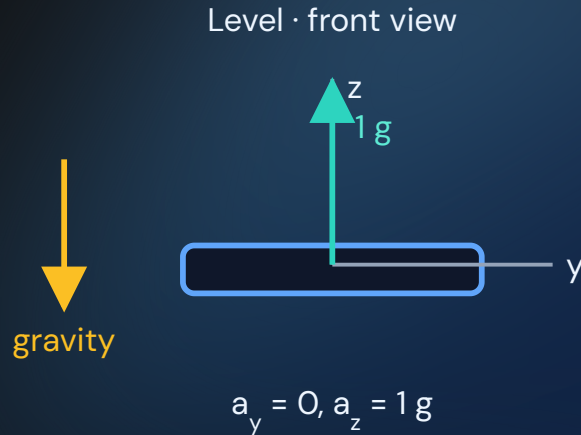
A tiny suspended mass deflects as the sensor is pushed. Electronics measure the deflection along three axes.

It measures **specific force**, in m/s^2 or g . At rest, its readings reveal the direction of gravity.

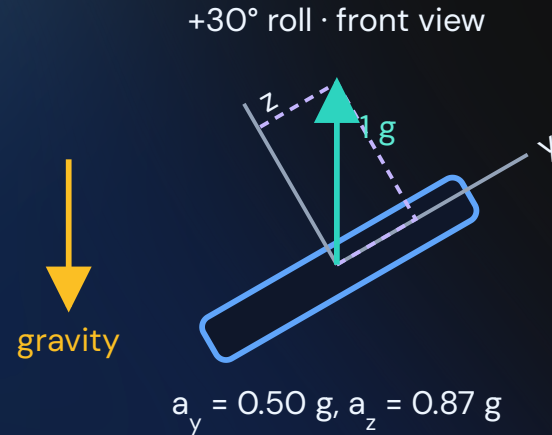
$1\text{ g} \approx 9.81\text{ m/s}^2$. A resting sensor reads about 1 g in total, not zero.



Tilt redistributes the gravity signal



Level: nearly all the stationary reading lies on z.



Rolled 30°: the same 1 g signal has both y and z components.

The robot's axes rotate; gravity stays vertical. The components tell us the tilt.

From acceleration to roll and pitch

ROLL · Φ · ROTATION ABOUT X

$$\phi = \text{atan2}(a_y, a_z)$$

For (0, 0.50g, 0.87g): **roll** $\approx 30^\circ$.

PITCH · Θ · ROTATION ABOUT Y

$$\theta = \text{atan2}\left(-a_x, \sqrt{a_y^2 + a_z^2}\right)$$

For (-0.50g, 0, 0.87g): **pitch** $\approx 30^\circ$.

Convention: x forward, y left, z up; level $a_z = +g$. Right-hand rotations; atan2 returns radians.

Works when gravity dominates: hold still or move gently. Driving acceleration and vibration distort tilt estimates.

Gravity alone cannot reveal yaw: turning on a level table leaves $(a_x, a_y, a_z) \approx (0, 0, g)$.

Magnetometer: a digital compass

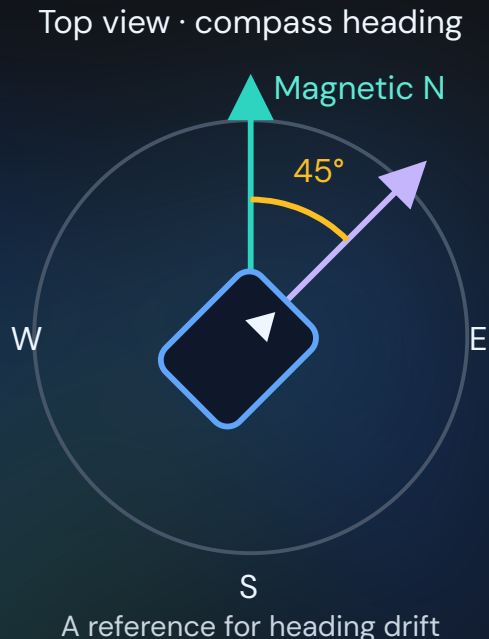
MAGNETIC FIELD · m_x, m_y, m_z

Measures Earth's magnetic field along three axes. Its horizontal direction provides a reference for **yaw**.

Yaw is rotation about the vertical axis: which way the robot faces.

Use roll and pitch to compensate for tilt when the robot is not level.

Calibrate the compass. Motors, magnets, and nearby steel can distort heading.



An IMU combines complementary senses

GYROSCOPE

Fast rotation changes

Integrate rates to track orientation. Small errors accumulate.

ACCELEROMETER

Gravity reference

Correct roll and pitch drift when gravity dominates.

MAGNETOMETER

Heading reference

Correct yaw drift when the magnetic field is reliable.

Sensor fusion combines these measurements into a better orientation estimate.

6-axis IMU: gyro + accelerometer. **9-axis package:** adds a magnetometer for heading.

Two sensor types track motion and tilt; the third anchors yaw. An IMU alone does not give drift-free position on a map.

Ultrasonic vs. IMU

ULTRASONIC

Distance in cm

- Uses sound + echo timing
- Changes near objects
- Looks primarily forward

IMU HEADING

Angle in degrees

- Uses internal motion sensing
- Changes when the robot turns
- Moves with the robot

Estimating position: what do we know?

1 · WHEEL VOLTAGE + TIME

Predict motion

Use a model to turn voltage into expected wheel speed, then speed $\times$ time into travel.

Weakness: load, battery, or a stalled wheel can make actual motion differ.

2 · WHEEL ENCODERS

Measure wheel travel

Convert measured wheel rotation into distance. Combine both wheels to estimate the path.

Weakness: wheel slip and wheel-size errors accumulate in the position estimate.

3 · RANGE FINDER + IMU

Relate motion to the world

Measure distance to a wall or landmark and estimate which way the robot faces.

Weakness: one range + heading does not uniquely locate the robot in 2D.

Toward a known wall: range constrains distance to the wall; IMU supplies orientation. Add landmarks or odometry to track the full path.

Sense the world. Sense yourself.

Distance supports obstacle avoidance. Heading supports precise turns. Today, build intuition for both.